

Mapping the UK Climate & Nature Policy Landscape for a new Met Office UK Land Surface Temperature (LST) product

Gap assessment · Prepared June 2026

Key points

- **This product does not yet exist at the Met Office.** This is a gap assessment: the external literature shows land surface temperature (LST) is used effectively elsewhere (urban heat islands, tree-canopy cooling, drought/evapotranspiration), and this asks what UK policy could do with a comparable MO product.
- **Key distinction:** satellite *LST* (skin temperature) is not *near-surface air temperature*. Almost all current UK heat policy — Heat-Health Alerts, overheating standards, CCRA/NAP heat risks — runs on air temperature, so direct LST uptake today is thin. That is precisely the gap.
- **Highest-value prospective applications (Relevance 3):** intra-urban heat-exposure mapping for the Weather-Health Alerting System and Adverse Weather and Health Plan, for CCRA4 / NAP4, and for planning under the NPPF (overheating, urban heat islands, green-infrastructure siting).
- **Timely:** UKHSA and the Met Office are already moving the Weather-Health Alerts toward finer geographic resolution — the natural insertion point for an intra-urban LST exposure layer.
- **Maturity varies:** urban green-infrastructure cooling is **Demonstrated** (tree-canopy LST studies); most heat and drought applications are **Emerging** (proven in research, not yet in UK policy); planning, overheating standards and ecosystem-condition uses are **Conceptual**.
- **Caveats:** the supplied bibliography is an extreme-LST topic reading list, not an MO publication list, so MO capability was scoped from MO/ESA sources; and four of the highest-relevance heat activities are operational processes not held in the governance workbook (marked †).

The evidence base for a prospective UK LST product

A UK LST product would deliver satellite-derived skin-temperature fields at fine spatial resolution — resolving intra-urban hot-spots, surface temperature extremes and surface-moisture stress in ways gridded air-temperature products cannot. The Met Office sits in a climate-services / model-evaluation role on the **ESA CCI Land Surface Temperature** record (led by the University of Leicester) and runs urban land-surface modelling (MORUSES / JULES) and Urban Climate Services. The external evidence base below shows the policy-relevant applications are established science elsewhere.

- **Urban heat island & exposure** — LST resolves within-city temperature variation that drives differential heat exposure and mortality (Zhao et al. 2021; Tuholske et al. 2021).
- **Green-infrastructure cooling** — tree-canopy and green/blue space measurably lower summer LST in temperate cities, quantifiable from LST (Wang, Scott & Dallimer 2023).
- **Surface temperature extremes** — LST captures land-surface maxima well above air temperature, relevant to infrastructure and ecosystem heat stress (Mildrexler et al. 2011; Garratt 1992).
- **Surface moisture / drought** — LST is a proxy for evaporative stress and soil-moisture deficit during heatwaves (Aminzadeh & Or 2014).

A prospective MO UK LST product would translate these demonstrated capabilities into a national, policy-ready evidence layer; the table maps where that would matter and how proven each use is.

Policy applications, relevance and maturity

Two independent axes. **Relevance** (materiality of a UK LST product to the policy): **3 High** = would be a direct, central input; **2 Medium** = a useful supporting evidence layer; **1 Low** = marginal or indirect. **Maturity** (how proven the application is): Demonstrated = proven elsewhere and transferable; Emerging = partial evidence, needs UK development/validation; Conceptual = plausible but unproven for this use. Items marked † are activities not held in the governance workbook. Policy names link to the source page.

Ref	Policy activity	Relevance	Maturity	What a UK LST product could enable for this policy
†	Heat-Health Alerting System / Weather-Health Alerts (UKHSA – Met Office)	3 HIGH	Emerging	Alerts run on Met Office near-surface air-temperature forecasts, not satellite LST. An LST product could add intra-urban hot-spot granularity to target interventions within alert regions — and UKHSA/Met Office are already moving toward finer-resolution alerts, the natural insertion point.
†	Adverse Weather and Health Plan (AWHP) (UKHSA)	3 HIGH	Emerging	The AWHP (a NAP commitment) is underpinned by the Weather-Health Alerting System and drives local heat-health planning. LST-derived intra-urban exposure surfaces could sharpen where warm-space and vulnerable-population actions are targeted; not yet an input.
POL-017	Climate Change Risk Assessment (CCRA3 2022; CCRA4 from 2026–27)	3 HIGH	Emerging	Heat risks to health, buildings and infrastructure are central CCRA risks. An LST product could quantify urban heat-island exposure for CCRA4 — a high-value, near-term insertion point — though CCRA presently uses air-temperature evidence.
POL-050	National Planning Policy Framework (NPPF)	3 HIGH	Emerging	The operative instrument for overheating, urban heat islands and green-infrastructure provision in local plans and decisions. LST-derived urban-heat surfaces could underpin heat-risk and cooling policies in plan-making; demonstrated in some local plans (e.g. London), not standardised nationally.
POL-002	National Adaptation Programme (NAP3) 2023–28	2 MED	Emerging	Translating CCRA heat risks into action would benefit from spatially explicit surface-heat exposure to prioritise cooling and overheating measures; the LST evidence exists but is not yet a NAP input.
POL-001	Environmental Improvement Plan 2025 (urban cooling / green infrastructure)	2 MED	Demonstrated	Green-infrastructure cooling is the best-demonstrated LST application: canopy and green/blue space measurably reduce surface temperature. An LST product would provide the national evidence layer for urban greening and cooling targets.
POL-010	England Peat Action Plan 2021	2 MED	Emerging	Peat surface temperature and moisture are proxies for condition, dryness and wildfire/burning risk on protected peatland. LST from thermal remote sensing is an emerging monitoring input for restoration targeting and burning-ban compliance.
†	Building Regulations Part O (overheating in new residential buildings)	2 MED	Conceptual	Part O assessments run on CIBSE design summer-year weather files (air temperature). An LST product could inform location-specific urban-heat uplift to those files — an inferred, not-yet-established pathway.
†	NSWWS Extreme Heat warnings (Met Office)	2 MED	Conceptual	Extreme-heat warnings are air-temperature and impact based; LST could add surface-heat context for built-up areas, but this is a speculative enhancement rather than a defined need.
POL-026	Natural Capital & Ecosystem Assessment (NCEA) Programme	2 MED	Conceptual	NCEA builds national baselines of ecosystem extent and condition. Surface temperature could feed a heat-stress condition layer for habitats and urban green space, but its use in condition accounts is not yet defined.
POL-009	Land Use Framework 2025	2 MED	Conceptual	Spatial land-use and development decisions could use LST-derived urban-heat surfaces to steer cooling-sensitive siting and green-space placement; plausible but not yet an articulated function.
POL-013	Environmental Land Management schemes (drought / soil-moisture stress)	1 LOW	Emerging	LST is a recognised proxy for evaporative stress and soil-moisture deficit; an LST product could support drought-sensitive land-management targeting, with research evidence but no current scheme use.
POL-058	Clean Air Strategy 2019 (heat – air-quality interaction)	1 LOW	Conceptual	Urban heat and ground-level ozone episodes coincide during heatwaves; LST urban-heat surfaces could contextualise air-quality risk in cities, but the link to this strategy is indirect.

Policy indicators

A scan of *biodiversity_indicators_manual_v6* for indicators whose measured quantity a UK LST product could directly inform — surface-temperature extremes and evaporative stress / drought — complemented by a focused web scan for authoritative indicators absent from the workbook. Pure GIS-area indicators (e.g. urban tree cover) were excluded because their measured quantity takes no climate input. These are supporting evidence links, consistent with LST's Emerging maturity for most UK uses. Items marked † are not held in the indicators workbook.

Indicator (ref)	Indicator name	Source framework	Current policy application(s)
IND-E-044	Disruption or unwanted impacts caused by drought	Environmental Indicator Framework (EIF)	EIP 2025 (Environment Act 2021); GBF
IND-C-145	Drought plan trigger levels and water restrictions incidence (England and Wales)	CCC Adaptation Monitoring Framework 2023–25	NAP3 / CCRA (Climate Change Act 2008, s.58)
IND-C-114	Forest and woodland resilience to climate change (drought, disease, wind) (UK)	CCC Adaptation Monitoring Framework 2023–25	NAP3 / CCRA (Climate Change Act 2008, s.58)
IND-C-130	Heat-related mortality and morbidity (England and Wales) — LST informs the intra-urban heat-exposure term, not the outcome count	CCC Adaptation Monitoring Framework 2023–25	NAP3 / CCRA (Climate Change Act 2008, s.58)
†	Heat mortality monitoring report, England — UKHSA Official Statistic (annual since 2016). LST could refine the intra-urban heat-exposure term (currently regional mean air temperature).	UKHSA Official Statistics (web scan; not in workbook — overlaps CCC IND-C-130)	Adverse Weather and Health Plan ; NAP3 Goals 1–2

Methods

- Capability scoped from external evidence**, not from MO outputs: the supplied bibliography is an extreme-LST topic reading list, used to establish what LST applications are demonstrated elsewhere and how mature each is for UK policy use.
- MO/ESA capability characterised** by inspecting MO and ESA pages (accessed June 2026): the ESA CCI LST_cci project (Met Office climate-services / CMUG partner), UKCP18 urban-resolving projections, the State of the UK Climate (NCIC) monitoring and attribution, and the Urban Climate Services team.
- Policy candidates** drawn from a full sweep of the governance workbook's Policies & Activities sheet (uk_climate_nature_governance_v15, 64 records; POL IDs stable across versions): screened across the Climate, Urban & Buildings, Health, Nature, Land and Water sectors for a plausible surface-temperature dependency. Supplemented with operational heat activities absent from the workbook — the Weather-Health Alerting System, the Adverse Weather and Health Plan, NSWWS Extreme Heat warnings and Building Regulations Part O — marked †. Marine (sea-surface-temperature) records were excluded as out of scope for a land product.
- Scoring** used two independent axes. *Relevance* (materiality of a UK LST product to the policy): central input = 3; supporting evidence layer = 2; marginal/indirect = 1. *Maturity*: Demonstrated where the application is proven elsewhere and transferable; Emerging where partial evidence needs UK development; Conceptual where the use is plausible but unproven. Kept separate so a high-value-but-immature application is not mistaken for a low-relevance one.
- Indicator scan** of *biodiversity_indicators_manual_v6* across its framework sheets, keyword-filtered for heat, surface/skin temperature and drought, then narrowed to indicators whose measured quantity an LST product could directly inform (surface-temperature extremes, evaporative stress / drought); urban tree-cover indicators were excluded because their measured quantity (a GIS area) takes no climate input. A focused web scan for authoritative indicators absent from the workbook found the UKHSA Heat mortality monitoring report (Official Statistic; overlaps CCC IND-C-130), marked †. Throughout, satellite LST (skin temperature) was treated as distinct from near-surface air temperature.

Main gaps and caveats

Two structural caveats. First, the bibliography is an external extreme-LST reading list, not an MO publication list, so MO capability was scoped from MO/ESA sources rather than the file. Second, satellite LST and near-surface air temperature are distinct: most UK heat policy runs on air temperature, so direct LST uptake is currently thin and most LST links here are Emerging or Conceptual. Four of the highest-relevance heat activities are operational processes not held in the governance workbook — the Weather-Health Alerting System, the Adverse Weather and Health Plan, NSWWS Extreme Heat warnings and Building Regulations Part O (marked †) — and are arguably worth adding as records. The Part O link runs via CIBSE weather files (inferred,

medium confidence); the NPPF, NCEA and Clean Air links are credible but not yet evidenced in current policy text. Sea-surface-temperature and marine-heatwave uses were excluded as out of scope for a land product.

References

Evidence base (external literature)

6. Aminzadeh, M. & Or, D. (2014) Soil evaporation and surface temperature during heat waves. *Water Resources Research*.
7. Mildrexler, D.J., Zhao, M. & Running, S.W. (2011) A global review of positive and negative land surface temperature extremes. *Scientific Reports*.
8. Garratt, J.R. (1992) Extreme maximum land surface temperatures. *Journal of Applied Meteorology*.
9. Tuholske, C. et al. (2021) Global urban population exposure to extreme heat. *Environmental Research Letters*.
10. Zhao, L. et al. (2021) Strong contributions of local background climate to the urban heat island effect. *Nature Climate Change*.
11. Wang, X., Scott, C.E. & Dallimer, M. (2023) High summer land surface temperatures in a temperate city are mitigated by tree canopy cover. *Urban Climate*.

Met Office / ESA capability and other sources

- ESA Climate Change Initiative — Land Surface Temperature (LST_cci); led by University of Leicester; Met Office climate-services / CMUG partner. climate.esa.int/.../land-surface-temperature
- Met Office (28 July 2022) Record high temperatures verified — 40.3°C at Coningsby, Lincolnshire.
- Kendon, M. et al. (2023) State of the UK Climate 2022 (Met Office National Climate Information Centre).
- World Weather Attribution (2022) Without human-caused climate change temperatures of 40°C in the UK would have been extremely unlikely.
- Met Office, Urban climate impacts research (MORUSES / JULES urban land-surface modelling) and Urban Climate Services 'Heat Packs'.
- UKHSA & Met Office, [Heat-Health Alerting system](#) / Weather-Health Alerts and [Adverse Weather and Health Plan](#) (GOV.UK, accessed June 2026).
- uk_climate_nature_governance_v15.xlsx, Policies & Activities sheet.